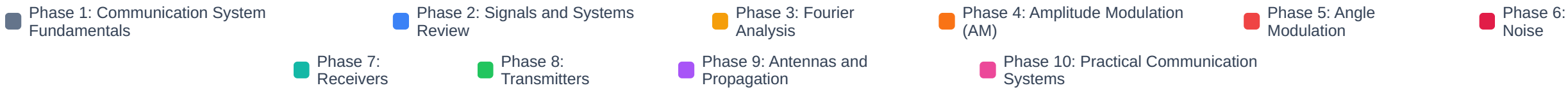


ANALOG COMMUNICATION

Complete Learning Roadmap — Beginner to Interview-Ready | 10 Phases · Prerequisite Flow · Milestones



PHASE 1

Communication System Fundamentals

KEY CONCEPTS

- Block diagram of a complete communication system
- Role of transmitter, channel, and receiver
- Analog vs digital communication basics
- Bandwidth, noise, interference & distortion

FULL TOPIC LIST

Communication elements (source, transducer, transmitter, channel, receiver, destination), Analog vs Digital, EM spectrum, frequency, wavelength, bandwidth, signal, message, noise, interference, distortion



Learning Outcome / Milestone: Explain the complete communication system using a block diagram.



PHASE 2

Signals and Systems Review

↳ Prerequisite refresher for all modulation math ahead

KEY CONCEPTS

- Classifying CT/DT and periodic/non-periodic signals
- Standard test signals: step, impulse, ramp, sinusoid
- Amplitude, frequency & phase of a waveform
- Energy vs power signal distinction

FULL TOPIC LIST

CT/DT, periodic/non-periodic, energy/power, even/odd signals; unit step, impulse, ramp, sinusoidal, exponential; amplitude, frequency, phase, time period



Learning Outcome / Milestone: Classify and sketch different signal types.



PHASE 3

Fourier Analysis

↳ Requires: signal classification (Phase 2)

KEY CONCEPTS

- Time-domain to frequency-domain conversion
- Frequency spectrum & harmonics of a signal
- Parseval's theorem — energy conservation
- Why every modulation scheme is analyzed in frequency domain

FULL TOPIC LIST

Fourier Series, Fourier Transform, frequency spectrum, harmonics, bandwidth, Parseval's theorem; applications in signal analysis, communication, filtering



Learning Outcome / Milestone: Convert a time-domain signal into the frequency domain.



PHASE 4

Amplitude Modulation (AM)

↳ Requires: Fourier analysis (Phase 3) to view AM spectra

KEY CONCEPTS

- Why modulation is needed for transmission
- AM principle & modulation index
- Power distribution across carrier and sidebands
- Comparing DSBFC, DSBSC, SSB and VSB

FULL TOPIC LIST

Need for modulation, carrier & message signal, AM principle, modulation index, power relations, frequency spectrum, bandwidth, generation & detection; types: Conventional AM, DSBFC, DSBSC, SSB, VSB



Learning Outcome / Milestone: Analyze and compare all AM techniques.



PHASE 5

Angle Modulation

↳ Requires: AM concepts (Phase 4) for comparison

KEY CONCEPTS

- FM vs PM principle & modulation index
- Frequency deviation and Carson's Rule for bandwidth
- Narrowband vs wideband FM
- FM/PM generation and detection circuits

FULL TOPIC LIST

FM, PM, modulation index, frequency deviation, Carson's Rule, narrowband/wideband FM, FM & PM generation and detection; AM vs FM vs PM comparison



Learning Outcome / Milestone: Explain why FM provides better noise immunity.





PHASE 6

Noise

↳ Explains the noise-immunity claims made in Phase 5

KEY CONCEPTS

- Internal vs external noise sources
- Thermal noise and shot noise mechanisms
- Signal-to-Noise Ratio (SNR) calculation
- Noise figure and noise temperature of a receiver

FULL TOPIC LIST

Internal & external noise, thermal noise, shot noise, atmospheric noise, industrial noise, SNR, noise figure, noise temperature



Learning Outcome / Milestone: Calculate SNR and identify different noise sources.



PHASE 7

Receivers

↳ Requires: modulation (Phase 4/5) + noise (Phase 6)

KEY CONCEPTS

- Superheterodyne receiver block diagram
- Role of mixer, local oscillator & IF stage
- Detector and AGC function
- TRF vs Superheterodyne trade-offs

FULL TOPIC LIST

TRF receiver, Superheterodyne receiver, RF amplifier, mixer, local oscillator, IF, detector, audio amplifier, AGC; TRF vs Superheterodyne comparison



Learning Outcome / Milestone: Draw and explain a Superheterodyne Receiver.



PHASE 8

Transmitters

↳ Requires: modulation techniques (Phase 4/5)

KEY CONCEPTS

- AM transmitter signal chain
- FM transmitter signal chain
- Oscillators & frequency synthesizers
- RF power amplifier stage

FULL TOPIC LIST

AM transmitter, FM transmitter, oscillators, RF power amplifier, frequency synthesizer



Learning Outcome / Milestone: Understand the signal flow in a transmitter.



PHASE 9

Antennas and Propagation

↳ Requires: transmitter/receiver RF stages (Phase 7/8)

KEY CONCEPTS

- Radiation from an antenna & radiation pattern
- Dipole vs monopole antenna basics
- Gain, directivity & polarization
- Ground wave, sky wave & space wave propagation

FULL TOPIC LIST

EM waves, radiation, antenna basics, dipole & monopole antenna, radiation pattern, gain, directivity, polarization; propagation: ground wave, sky wave, space wave



Learning Outcome / Milestone: Select suitable antennas for different communication systems.



PHASE 10

Practical Communication Systems

↳ Integrates: all previous phases into applied systems

KEY CONCEPTS

- AM/FM radio and TV broadcasting systems
- Aircraft & marine communication basics
- Satellite communication fundamentals
- Wireless communication system overview

FULL TOPIC LIST

AM radio, FM radio, television broadcasting, aircraft communication, marine communication, satellite communication basics, wireless communication basics



Learning Outcome / Milestone: Explain how analog communication is used in real-world systems.

AM vs FM vs PM

Quick comparison of the three core modulation schemes covered in Phases 4 & 5

AM

Amplitude varies with message; constant frequency

Narrow bandwidth, simple receivers, low noise immunity

FM

Frequency varies with message; constant amplitude

Wider bandwidth (Carson's Rule), strong noise immunity

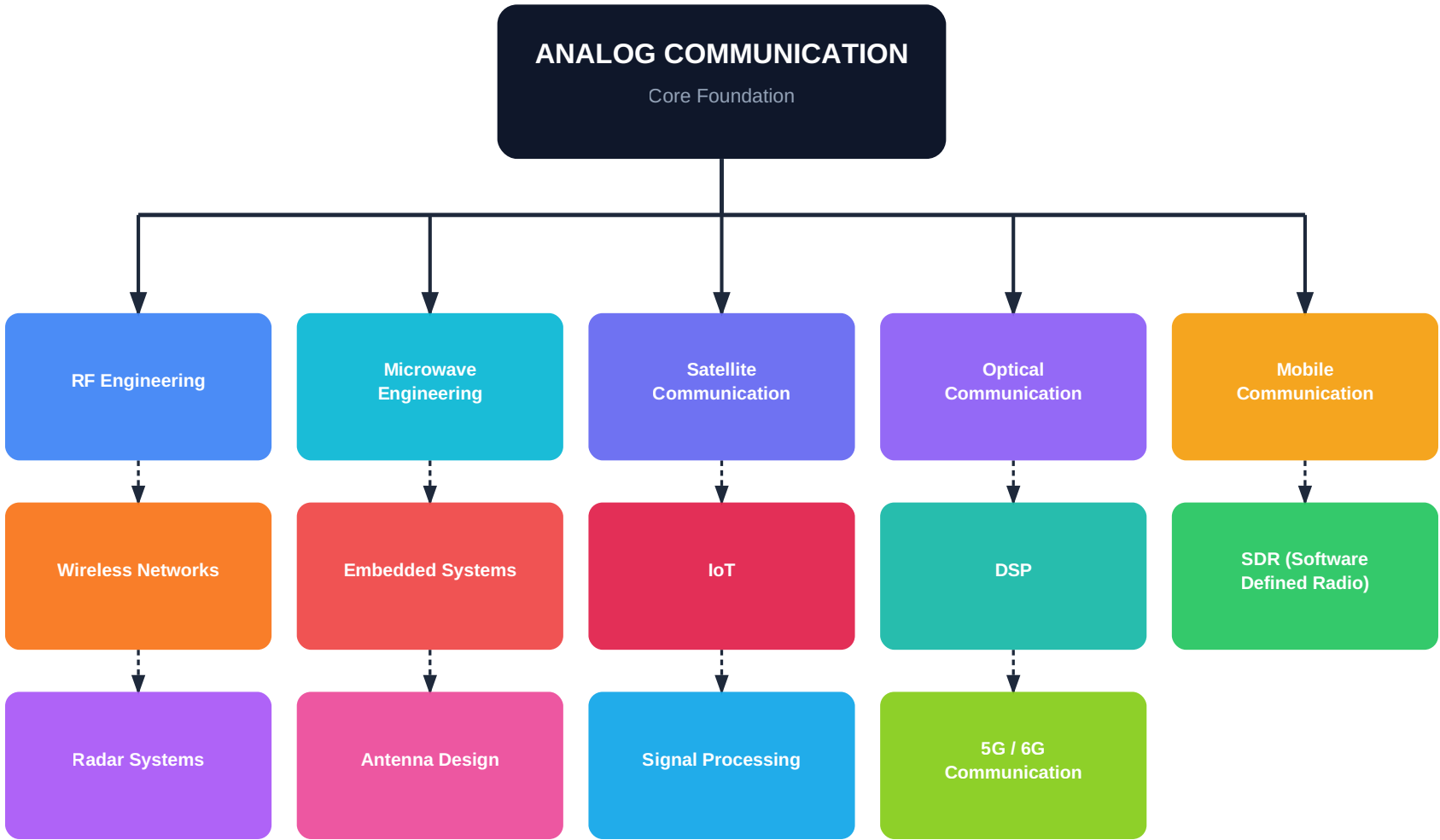
PM

Phase varies with message; constant amplitude

Closely related to FM, used in digital & hybrid systems

Final Connections — Where Analog Communication Leads

How this foundation branches into advanced ECE subjects and industry domains



The Learning Cycle

Repeat this 10-step workflow for every topic in the roadmap above

